

Black Hole Problems

1. At the center of the Milky Way lies a supermassive blackhole by the name of Sagittarius A*. The mass of Sagittarius A*, M , is estimated to be

$$M \approx 8.25 \times 10^{36} \text{ kg.}$$

The Schwarzschild radius of a black hole, r_s , is related to its mass M by

$$r_s = \frac{2GM}{c^2}.$$

What is the Schwarzschild radius of Sagittarius A* in meters (m)? Round your answer to two decimal places in scientific notation. Reminder: keep this task card handy, you will need it later!

2. Dhruv's planet orbits the black hole TON-430 at a constant distance $R = 2.1 \times 10^{10}$ m from the singularity. TON-430 has a mass $M = 3.0 \times 10^{36}$ kg. The velocity of a satellite orbiting a body of mass M at a distance R satisfies the relation

$$v^2 - \frac{GM}{R} = 0.$$

What is the orbital velocity of Dhruv's planet? Round your answer to two decimal places in scientific notation.

3. The local gravitational acceleration on the surface of the Earth is known to be $-g_{\text{earth}}$ where $g_{\text{earth}} = 9.8 \text{ m/s}^2$.

However, the general formula for the local gravitational acceleration at a distance R away from a body of mass M is given by

$$g = \frac{GM}{R^2}.$$

Using the first task card, find the local gravitational acceleration of a body at a distance of a Schwarzschild radius away from the singularity of Sagittarius A*. Round your answer to two decimal places in scientific notation.

4. The Hawking temperature is the theoretical temperature of the thermal radiation emitted by a black hole near its event horizon. Its formula is given by

$$T_{\text{hawking}} = \frac{\hbar c^3}{8\pi G M k_b}$$

where T is in Kelvin. What is the absolute difference of the Hawking temperature between Sagittarius A* and TON-430? Round your answer to two decimal places in scientific notation. Hint: You will need the first and second task cards, as well as the table of constants!

5. When objects veer near black holes, they experience time slower! The formula for time dilation near a black hole is given by

$$t' = t \sqrt{1 - \frac{r_s}{R}}.$$

Here, t' is the experienced time, t is the normal time measured far from the black hole, r_s is the Schwarzschild radius of the black hole, and R is the distance from the black hole. For every hour experienced on Earth (far from TON-430), how much time has passed on Dhruv's planet (refer to the second task card!)? Round your answer to two decimal places.

6. The density of a black hole ρ is related to its Schwarzschild radius r_s and can be approximated as

$$\rho \approx \frac{3M}{4\pi r_s^3}.$$

- (a) Using the approximation, find the density of a black hole ρ *only in terms* of its mass M and relevant constants (G , c , π). Hint: refer back to the first task card!
- (b) Using part (a), find the density of TON-430 using its given mass M (refer to the second task card!). Round your answer to two decimal places in scientific notation.